SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
   H H T H H H T H H T
   Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

Maximum Likelihood Estimation (MLE)

Given:

- Total number of coin tosses, **n = 10**
- Outcomes: **H H T H H H T H H T**

Step 1: Count the number of Heads and Tails

- Number of Heads (H) = **7**
- Number of Tails (T) = **3**

Step 2: Apply the MLE Formula

For a Bernoulli (coin toss) experiment, the Maximum Likelihood Estimate of the probability of getting Heads is:

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Number of Tosses}}$$

Step 3: Calculate

$$\hat{p} = \frac{7}{10} = 0.7$$

Final Answer

- **Estimated probability of getting Heads (p): 0.7**
- **Estimated probability of getting Tails (1 − p): 0.3**

Conclusion
Using **Maximum Likelihood Estimation (MLE)**, the estimated probability of getting **Heads** is:

$$\boxed{\hat{p} = 0.7}$$

Hence, the MLE estimate of the probability of getting **Heads is 0.7 (or 70%)**.

2. Out of 50 students, 42 passed an exam.
   Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Maximum Likelihood Estimation (MLE)
Given:

- Total number of students, **n = 50**
- Number of students who passed = **42**
- Number of students who failed = **50 − 42 = 8**

Step 1: Apply the MLE Formula
For a Bernoulli distribution, the Maximum Likelihood Estimate (MLE) of the probability of success (passing) is:

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Number of Trials}}$$

Step 2: Calculate

$$\hat{p} = \frac{42}{50} = 0.84$$

Final Answer

- **MLE of the probability that a student passes:**

$$\boxed{\hat{p} = 0.84}$$

- **Probability of failure:**

$$1 - \hat{p} = 1 - 0.84 = 0.16$$

Conclusion
Using **Maximum Likelihood Estimation (MLE)**, the estimated probability that a student **passes the exam** is:

$$\boxed{0.84 \text{ (or } 84\%)}$$

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
   Find the MLE for the probability that a bulb is defective.

Maximum Likelihood Estimation (MLE)

Given:

- Total number of bulbs, **n = 100**
- Number of defective bulbs = **8**
- Number of non-defective bulbs = **100 − 8 = 92**

Step 1: Apply the MLE Formula

For a Bernoulli distribution, the Maximum Likelihood Estimate (MLE) of the probability of success is:

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Number of Trials}}$$

Here, a **defective bulb** is considered the success event.

Step 2: Calculate

$$\hat{p} = \frac{8}{100} = 0.08$$

Final Answer

- **MLE of the probability that a bulb is defective:**

$$\boxed{\hat{p} = 0.08}$$

- **Probability that a bulb is non-defective:**

$$1 - \hat{p} = 1 - 0.08 = 0.92$$

Conclusion

Using **Maximum Likelihood Estimation (MLE)**, the estimated probability that a bulb is **defective** is:

$$\boxed{0.08 \text{ (or } 8\%)}$$

4. A biased die is rolled 60 times. The number 6 appears 18 times. Estimate the probability of rolling a 6 using MLE.

Maximum Likelihood Estimation (MLE)
Given:

- Total number of die rolls, **n = 60**
- Number of times the face **6** appears = **18**
- Number of times other faces appear = **60 − 18 = 42**

Step 1: Apply the MLE Formula

For a Bernoulli distribution, the Maximum Likelihood Estimate (MLE) of the probability of rolling a **6** is:

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Number of Trials}}$$

Here, rolling a **6** is considered the success event.

Step 2: Calculate

$$\hat{p} = \frac{18}{60} = 0.30$$

Final Answer

- **MLE of the probability of rolling a 6:**

$$\boxed{\hat{p} = 0.30}$$

- **Probability of not rolling a 6:**

$$1 - \hat{p} = 1 - 0.30 = 0.70$$

Conclusion

Using **Maximum Likelihood Estimation (MLE)**, the estimated probability of rolling a **6** is:

$$\boxed{0.30 \text{ (or } 30\%)}$$

5. Given:

   Input (x) = 4

   Weight (w) = 2

   Actual Output = 10

   Prediction:

   y = w × x

   1. Find the predicted output.
   2. Calculate the error.
   3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Given:

- Input, **x = 4**
- Weight, **w = 2**
- Actual Output = **10**
- Prediction Formula: **y = w × x**
- Gradient = **4**
- Learning Rate, **η = 0.01**

1. Find the Predicted Output

Using the prediction formula:

$$y = w \times x$$
$$y = 2 \times 4 = 8$$

Predicted Output = $\boxed{8}$

2. Calculate the Error

Error is calculated as:

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$
$$\text{Error} = 10 - 8 = 2$$

Error = $\boxed{2}$

3. Find the Updated Weight

Gradient Descent update rule:

$$w_{\text{new}} = w - \eta \times \text{Gradient}$$

Substitute the given values:

$$w_{\text{new}} = 2 - (0.01 \times 4)$$
$$w_{\text{new}} = 2 - 0.04 = 1.96$$

Updated Weight = $\boxed{1.96}$

Final Answers

1. **Predicted Output = 8**
2. **Error = 2**
3. **Updated Weight = 1.96**

Code of Conduct:

*I **M. Srilakshmi (192525404)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Srilakshmi

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation